

Računarske mreže

(20ER5003)

Lab. 6 – CGI

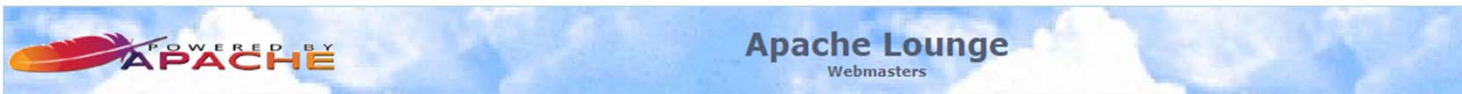
Priprema za lab. vežbu



Priprema za izvođenje vežbe

- Preuzeti neki Web server koji podržava CGI, recimo Apache
- Poželjno je preuzeti izbuildovanu verziju:
 - ▷ <https://www.apachelounge.com/download/>
- Raspakovati sadržaj
- Pokrenuti Apache server (Apache24\bin\httpd.exe)
- CGI aplikaciju postaviti u direktorijum \Apache24\cgi-bin\)
- Podrazumevani direktorijum za stranice je \Apache24\htdocs\

Apache 2.4 VS17 Windows Binaries



Home

VS17

VS16

Additional

NEW

18 December 2022
mod_wasm added

09 November 2022
New C++ Redistributable

08 November 2022
httpd 2.4.54 and modules
VS17 available

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Apache 2.4 VS17 Windows Binaries and Modules

Apache Lounge has provided up-to-date Windows binaries and popular third-party modules for more than 15 years. We have hundreds of thousands of satisfied users: small and big companies as well as home users. Always build with up to date dependencies and latest compilers, and tested thorough. The binaries are referenced by the ASF, Microsoft, PHP etc. and more and more software is packaged with our binaries and modules.

The binaries, are build with the sources from ASF at <http://httpd.apache.org>, contains the latest patches and latest dependencies like zlib, openssl etc. which makes the downloads here mostly more actual then downloads from other places. The binaries **do not run** on XP and 2003. Runs on: 7 SP1, Vista SP2, 8/8.1, 10, 11 Server 2008 SP2 / R2 SP1, Server 2012 / R2, Server 2016/2019/2022.

Build with the latest Windows® Visual Studio C++ 2022 aka VS17. Has improvements, fixes and optimizations over VC16 in areas like Performance, MemoryManagement, New standard conformance features, Code generation and Stability. For example code quality tuning and improvements done across different code generation areas for "speed". And makes more use of latest processors and supported Windows editions (win7 and up) internal features.

VS17 is backward compatible, That means, a VS16/15/14 module can be used inside the VS17 binary.

Be sure you installed latest 14.34.31931 Visual C++ Redistributable Visual Studio 2015-2022 : [vc_redist_x64](#) or [vc_redist_x86](#) see [Redistributable](#)

Apache 2.4 binaries VS17

Info & Changelog

Apache 2.4.54 Win64

• [httpd-2.4.54-win64-VS17.zip](#)

08 Nov '22 11.027K

PGP Signature (Public [PGP key](#)), SHA1-SHA512 [Checksums](#)

To be sure that a download is intact and has not been tampered with, use PGP, see [PGP Signature](#)

Apache 2.4 modules VS17

Mail for the PGP signatures and/or SHA checksums to verify the contents of a file.

Provera funkcionisanja

- Napraviti **konzolnu aplikaciju** u Visual Studio okruženju
- U **main** funkciji dodati kod za minimalnu CGI aplikaciju
- Dobijeni exe fajl prebaciti u cgi-bin direktorijum
- Pokrenuti aplikaciju iz browser-a (<http://localhost/cgi-bin/cgi.exe>)

```
#include <iostream>

int main()
{
    std::cout << "Content-Type: text/html\n\n";
    std::cout << "<html> <head>\n";
    std::cout << "<title>Hello, World!</title>\n";
    std::cout << "</head>\n";
    std::cout << "<body>\n";
    std::cout << "<h1>Hello, World!</h1>\n";
    std::cout << "</body> </html>\n";
}
```



Priprema na CS-u

RAČUNARSKE MREŽE

Laboratorijska vežba br.6 – CGI

Priprema

1. Napraviti HTML formu za unos dva broja (po jedno polje za broj) i željene operacije (SELECT tag sa opcijama +, - i *) i CGI aplikaciju koja primenjuje datu operaciju i vraća rezultat klijentu u obliku HTML stranice. Koristiti HTTP POST metodu za prosleđivanje parametara.
2. Napraviti HTML formu za unos jednog broja i CGI aplikaciju koja ukoliko je broj paran vraća jednu sliku, ukoliko je neparan drugu. Koristiti HTTP GET metodu za prosleđivanje parametara.
3. Napraviti HTML formu koja prikazuje sliku 1 i CGI aplikaciju koja vraća korisniku informaciju da li je kliknuo na žuti ili crveni deo slike.



Slika 1

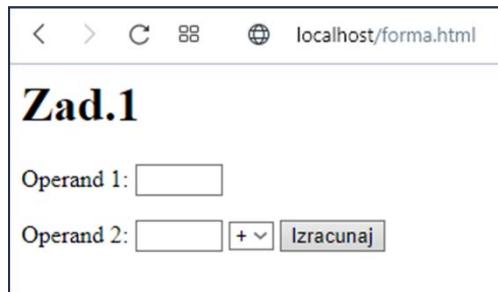
Ispravnost funkcionisanja može se proveriti na Small HTTP server-u. Formirati direktorijum cgi-bin ispod njegovog osnovnog direktorijuma (npr. c:\shhttps\www\cgi-bin\) i u njega smetiti odgovarajuću CGI aplikaciju i potrebne pomoćne datoteke. Prilikom pozivanja sa lokalnog računara koristiti loopback adresu 127.0.0.1.

Zadatak 1.

- Napraviti HTML formu za unos dva broja (po jedno polje za broj) i željene operacije (SELECT tag sa opcijama +, - i *) i CGI aplikaciju koja primenjuje datu operaciju i vraća rezultat klijentu u obliku HTML stranice. Koristiti HTTP POST metodu za prosleđivanje parametara.

Forma

```
<html> <head>
<title>Zad.1</title>
</head>
<body>
<h1>Zad.1</h1>
<form action="\cgi-bin\cgi.exe" method=POST>
<p>Operand 1: <input name="op1" size=3>
<p>Operand 2: <input name="op2" size=3>
<select name="operacija">
<option value="1">+</option>
<option value="2">-</option>
<option value="3">*</option>
</select>
<input type=submit value="Izracunaj">
</form>
</body> </html>
```



The screenshot shows a web browser window with the address bar displaying 'localhost/forma.html'. The page content includes a heading 'Zad.1' and a form with two input fields labeled 'Operand 1:' and 'Operand 2:'. The 'Operand 2:' field is followed by a dropdown menu with a '+' icon and a submit button labeled 'Izracunaj'.

CGI aplikacija

```
int main()
{
    std::string reqmethod = getenv("REQUEST_METHOD");
    std::string q_string = getenv("QUERY_STRING");
    std::string postInputString;

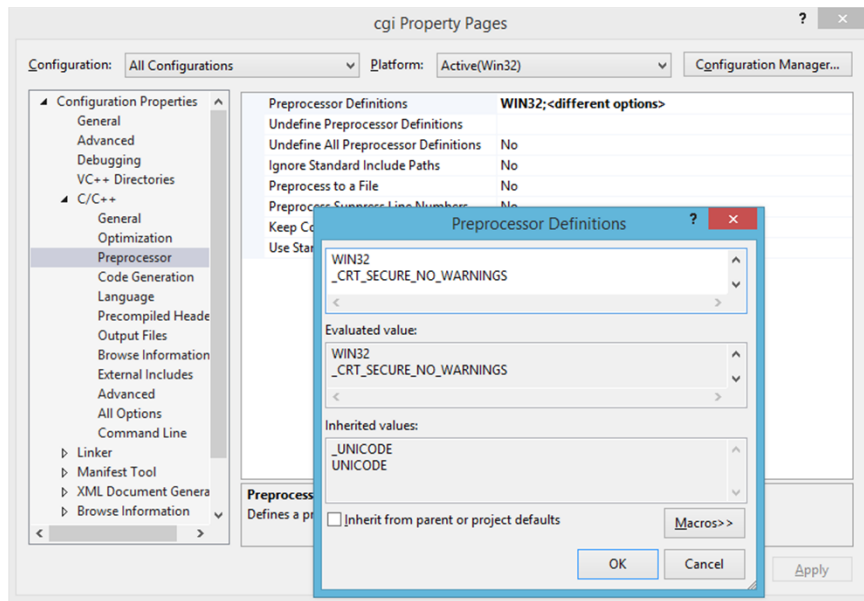
    if (reqmethod == "GET")    // GET zahtev
    {
        if (!q_string.empty())
            parse(q_string);
    }
    else if (reqmethod == "POST")    // POST zahtev
    {
        if (!q_string.empty())
            parse(q_string);

        int br_byte = atoi(getenv("CONTENT_LENGTH"));

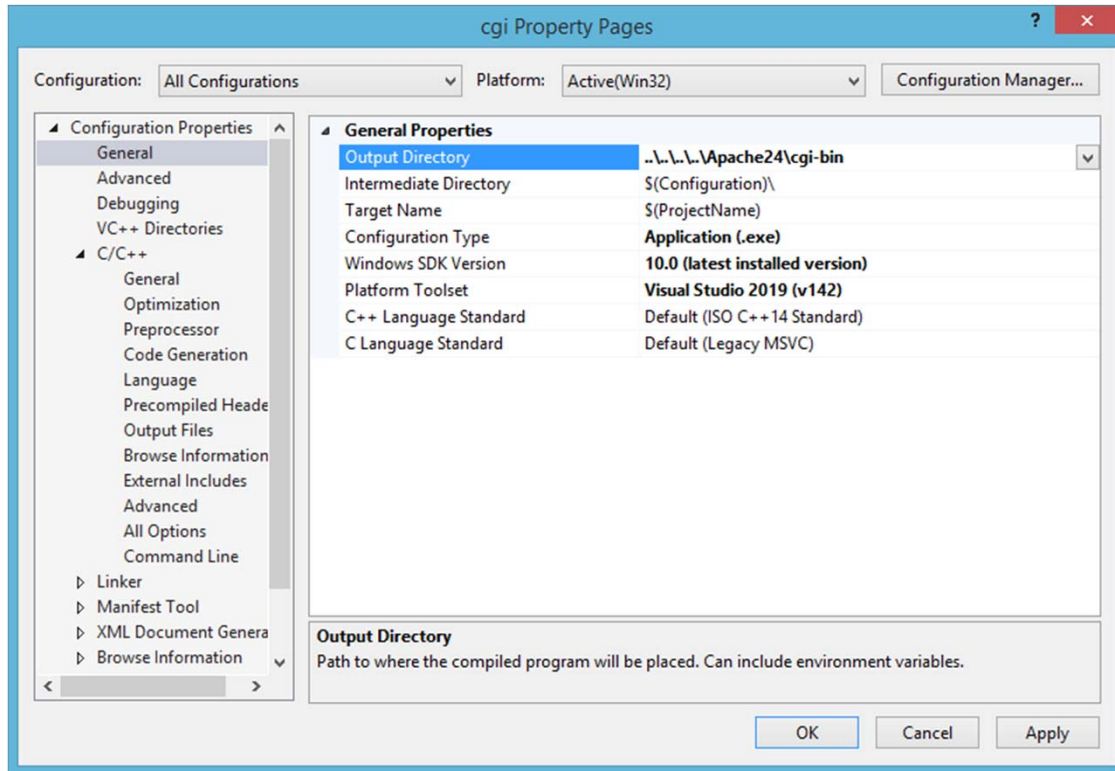
        if (br_byte > 0)
        {
            std::cin >> postInputString;
            parse(postInputString);
        }
    }
}
```

__CRT_SECURE_NO_WARNINGS

Error List				
Entire Solution		2 Errors	0 Warnings	0 Messages
Build + IntelliSense				
Code	Description	Project	File	
C4996	'getenv': This function or variable may be unsafe. Consider using _dupenv_s instead. To disable deprecation, use __CRT_SECURE_NO_WARNINGS. See online help for details.	cgi	cgi.cpp	
C4996	'getenv': This function or variable may be unsafe. Consider using _dupenv_s instead. To disable deprecation, use __CRT_SECURE_NO_WARNINGS. See online help for details.	cgi	cgi.cpp	



Output Directory



Parsiranje zahteva

QUERY_STRING: **op1=111&op2=22&operacija=3**

```
#include <iostream>
#include <string>

int result;

void parse(std::string str)
{
    int pos1 = str.find("op1=", 0);
    int pos2 = str.find("&", pos1);
    std::string val1 = str.substr(pos1 + 4, pos2 - pos1 - 4);
    int ival1 = std::stoi(val1);
}
```

Vraćanje rezultata

```
std::cout << "Content-Type: text/html\n\n";  
std::cout << "<html> <head>\n";  
std::cout << "<title>Zad.1</title>\n";  
std::cout << "</head>\n";  
std::cout << "<body>\n";  
std::cout << "<h1>Lab. vezba 6 - Zad.1</h1>\n";  
std::cout << "<p>Rezultat je: "<< result << "\n";  
std::cout << "</body> </html>\n";
```

localhost/forma.html

Zad.1

Operand 1:

Operand 2:

localhost/cgi-bin/cgi.exe

Lab. vezba 6 - Zad.1

Rezultat je: 2442

PITANJA

